

Physics Formula and Units Sheet

Mechanics, matter, waves and electricity

Use this sheet for revision and problem solving. Always write the known values with units, choose the relationship, substitute, calculate, and check whether the result is reasonable.

Clear explanations - consistent SI units - ready to print

Mechanics and matter

Relationship	Equation	SI result
Speed	$v = d / t$	m/s
Acceleration	$a = \text{change in } v / t$	m/s^2
Force	$F = ma$	N
Weight near Earth	$W = mg$	N
Momentum	$p = mv$	kg m/s
Kinetic energy	$E_k = \frac{1}{2} mv^2$	J
Potential energy	$E_p = mgh$	J
Work	$W = Fs$ (parallel force)	J
Power	$P = E / t$	W
Density	$\rho = m / V$	kg/m^3
Pressure	$p = F / A$	Pa

Useful constants

Near Earth's surface, g is commonly approximated as 9.81 m/s^2 . Use the value specified by the question when one is provided.

Waves and electricity

Relationship	Equation	SI result
Wave speed	$v = f \lambda$	m/s
Frequency and period	$f = 1 / T$	Hz
Ohm's law	$V = IR$	V
Electrical power	$P = VI$	W
Electrical energy	$E = Pt$	J
Charge flow	$Q = It$	C

Unit checklist

Convert all values into compatible units before substitution. A correct-looking number with an incompatible unit is not a correct scientific result.

Common prefixes

kilo (k) = 10^3 ; mega (M) = 10^6 ; milli (m) = 10^{-3} ; micro = 10^{-6} ; nano (n) = 10^{-9} .